

PERM, Russian Federation

WMO: 282240

Lat: 57.9925N

Lon: 56.3356E

Elev: 171

StdP: 99.29

Time Zone: 5.00 (E05)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.1	-25.8	-32.2	0.2	-28.9	-28.5	0.3	-25.5	7.4	-7.1	6.7	-7.1	1.2	10	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	29.6	20.2	27.8	19.2	25.9	18.1	21.2	27.8	20.2	26.1	19.3	24.4	2.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.0	23.9	18.1	13.3	22.9	17.2	12.5	22.3	62.1	28.1	58.9	26.1	55.6	24.3	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.7	5.1	-32.8	32.3	4.9	1.8	-36.4	33.6	-39.3	34.7	-42.0	35.8	-45.6	37.1		
			-32.9	22.4	4.9	1.0	-36.5	23.1	-39.4	23.7	-42.1	24.3	-45.7	25.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-13.2	-11.7	-4.0	4.2	11.6	16.2	18.4	16.4	10.6	3.2	-4.0	-10.0
	DBStd	12.30	7.22	6.54	4.83	4.53	5.20	4.46	4.31	4.24	4.10	4.62	6.02	7.22
	HDD10.0	3297	719	607	434	183	44	6	1	2	41	217	422	621
	HDD18.3	5641	978	841	692	425	215	92	54	88	234	470	672	879
	CDD10.0	821	0	0	0	8	95	192	261	201	58	5	0	0
	CDD18.3	123	0	0	0	1	8	27	57	29	2	0	0	0
	CDH23.3	1094	0	0	0	1	98	235	520	232	8	0	0	0
	CDH26.7	288	0	0	0	0	14	52	158	64	0	0	0	0
(14) Wind	WSAvg	2.4	2.5	2.4	2.8	2.7	2.5	2.2	2.0	1.9	2.2	2.6	2.7	2.5
(15) Precipitation	PrecAvg	614	39	26	27	35	54	70	79	74	63	60	48	39
	PrecMax	874	71	71	75	91	150	172	158	175	136	101	86	67
	PrecMin	398	9	4	0	3	14	9	16	24	12	10	17	3
	PrecStd	123	16	14	17	20	34	42	38	37	34	24	20	15
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	1.6	9.8	21.9	28.9	30.2	32.0	32.1	24.5	17.3	9.2	2.4
		MCWB	0.4	0.3	4.7	13.1	17.0	19.0	20.6	21.6	15.9	10.4	7.9	1.9
	2%	DB	0.2	0.6	6.5	16.9	26.2	28.1	30.1	28.0	21.2	14.0	6.9	1.2
		MCWB	-0.3	-0.4	2.4	10.3	15.5	18.8	21.1	20.2	14.7	9.7	5.7	0.6
	5%	DB	-1.4	-0.9	4.0	13.7	23.8	26.2	28.4	25.6	18.8	11.5	4.8	0.4
		MCWB	-2.0	-2.0	1.3	7.9	14.5	18.1	20.2	18.5	13.8	8.3	4.0	-0.2
	10%	DB	-3.5	-2.7	2.1	11.1	21.1	24.0	26.4	23.6	16.6	9.6	2.5	-0.9
		MCWB	-4.2	-3.9	0.2	6.2	13.0	17.2	19.3	17.7	13.1	7.6	1.6	-1.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	0.8	5.1	13.2	18.3	20.9	22.6	21.9	17.0	11.4	8.2	2.1
		MCDB	1.1	1.4	9.4	20.0	26.4	27.1	29.0	29.5	21.9	15.1	9.5	2.2
	2%	WB	-0.3	-0.3	3.0	10.8	16.5	19.9	21.6	20.7	15.6	10.1	5.8	0.7
		MCDB	0.1	0.6	5.1	15.6	24.3	25.9	28.4	27.0	19.6	13.2	6.7	1.0
	5%	WB	-1.9	-1.9	1.6	8.5	15.2	19.0	20.7	19.6	14.6	8.7	3.9	-0.2
		MCDB	-1.4	-0.9	3.5	12.5	22.2	24.8	26.9	24.5	17.5	10.9	4.8	0.4
	10%	WB	-4.1	-3.9	0.5	6.9	13.9	17.9	19.8	18.3	13.5	7.6	1.7	-1.7
		MCDB	-3.4	-2.9	2.0	10.6	19.9	23.1	25.3	22.4	16.1	9.6	2.3	-1.0
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	6.9	7.3	8.3	10.8	9.5	9.7	8.8	7.0	4.6	3.9	4.8
		MCDBR	4.9	5.2	8.7	12.1	14.1	12.5	12.5	11.8	9.9	8.0	4.8	3.5
	5% WB	MCWBR	4.7	4.4	5.8	6.9	6.2	5.3	5.0	5.2	4.9	4.8	4.4	3.6
		MCDWR	5.0	5.0	7.6	10.9	12.9	11.5	11.4	11.0	8.2	7.0	5.0	3.6
(40) Clear-Sky Solar Irradiance	taub	0.290	0.316	0.371	0.401	0.392	0.388	0.411	0.424	0.377	0.347	0.308	0.285	(40)
		1.999	2.006	1.938	2.086	2.302	2.369	2.311	2.247	2.341	2.274	2.095	1.970	(41)
	Edn at Noon	503	679	741	798	841	850	817	772	755	658	485	378	(42)
		59	93	133	136	117	112	116	116	90	72	54	45	(43)
(44) All-Sky Solar Radiation	RadAvg	0.52	1.39	2.66	3.95	5.05	5.37	5.41	4.02	2.37	1.01	0.43	0.28	(44)
	RadStd	0.08	0.23	0.34	0.54	0.56	0.47	0.57	0.42	0.23	0.14	0.09	0.06	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page